

Large-Scale Climate Variability and Mortality: An Exploratory Study of ENSO in Bangalore, Karnataka

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Abstract

This exploratory study examines the relationship between large-scale climate variability, represented by El Niño–Southern Oscillation (ENSO), and mortality rates in Bangalore, Karnataka. Data from various sources was aggregated, cleaned and transformed to build a dataset consisting of monthly, local weather variables and Oceanic Niño Index values spanning from 2001 to 2023. Multiple linear regression models were developed and compared to assess the effects of ENSO phases, seasonality, temperature and rainfall on mortality. The results indicate a more consistent association between ENSO phase and mortality whereas variables like rainfall and seasonality are not significant predictors. The findings suggest that large scale climate variability may have a greater impact on mortality when compared to local weather conditions. This pilot study demonstrates a framework for integrating climate and public health data and may be expanded to larger datasets, additional environmental variables and geographic locations.

Introduction

The El Niño–Southern Oscillation (ENSO) is one of the strongest naturally occurring causes of global climate variability. ENSO influences temperature, precipitation and various climatic conditions worldwide through changes in wind patterns and sea surface temperatures. These environmental changes have been linked to a variety of public health outcomes, including infectious disease transmission, heat-related illness, and mortality⁴.

India is particularly vulnerable to ENSO-related climate variability due to its location. Bangalore, a city in Karnataka, is impacted by this phenomenon, making it an important location for examining potential climate health relationships.

While previous studies have investigated the impacts of ENSO on disease outbreaks and climate patterns, few have examined the relationship between ENSO and all-cause mortality in India's cities. This study integrates mortality, weather, and Oceanic Niño Index (ONI) data from 2001–2023 to explore whether ENSO phase and local weather variables are associated with mortality in Bangalore using multiple linear regression methodology.

Data Collection and Processing

A monthly dataset was developed by integrating mortality and climate data for Bangalore. The final dataset contained 229 monthly observations spanning January 2001 to December 2023, excluding years with incomplete or confounding data (2006, 2020–2022, and 2024). Data were aggregated and processed using R to create a dataset containing mortality outcomes, local weather variables, and Oceanic Niño Index (ONI) values.

Mortality data were obtained from the Government of Karnataka's Annual Reports on the Registration of Births and Deaths Act, 1969, accessed through OpenCity. Monthly mortality counts and mid-year population estimates for Bengaluru Urban and Bengaluru Rural districts were transcribed from annual reports. Mortality rates were standardized per every 10,000 residents using the formula:

$$\text{Deaths per 10,000} = \frac{\text{Monthly Deaths}}{\text{Mid-Year Population}} * 10,000$$

Mortality rates were separately calculated for Bengaluru Urban and Bengaluru Rural districts, and a combined mortality rate was calculated encapsulating both districts.

Weather observations were obtained from the India Meteorological Department (IMD) through OpenCity. Daily weather observations were aggregated into monthly variables, with daily maximum temperatures averaged to calculate monthly average maximum temperature and daily rainfall measurements summed to calculate monthly total rainfall. The resulting monthly weather dataset was merged with mortality data by month.

The Oceanic Niño Index (ONI) represents ENSO variability, obtained from the National Oceanic and Atmospheric Administration (NOAA). The first monthly value from each three-month running mean was incorporated into the dataset. ONI values were categorized using R into the following ENSO phases:

$$\begin{aligned} \text{El Niño} &= ONI > 0.5 \\ \text{La Niña} &= ONI < -0.5 \\ \text{Neutral} &= -0.5 < ONI < 0.5 \end{aligned}$$

Data from 2020–2022 were excluded due to disruptions associated with the COVID-19 pandemic. 2024 observations were excluded due to incomplete weather records. Administrative boundary changes in 2007 were accounted for using updated district population estimates reported by the Government of Karnataka Annual Report.

Exploratory Data Analysis

Preliminary exploratory data analysis was conducted to analyze the dataset before developing a regression model. In order to identify changes in variables across the study period, time series visualizations were utilized.

The Oceanic Niño Index (ONI) demonstrated continuous variability between 2001 and 2023, reflecting the cyclical nature of ENSO events. Periods of positive ONI values corresponded with El Niño conditions, while negative ONI values corresponded with La Niña conditions. These fluctuations suggest an influence of climate variability. Hence, ENSO phase was used as a predictor in subsequent models.

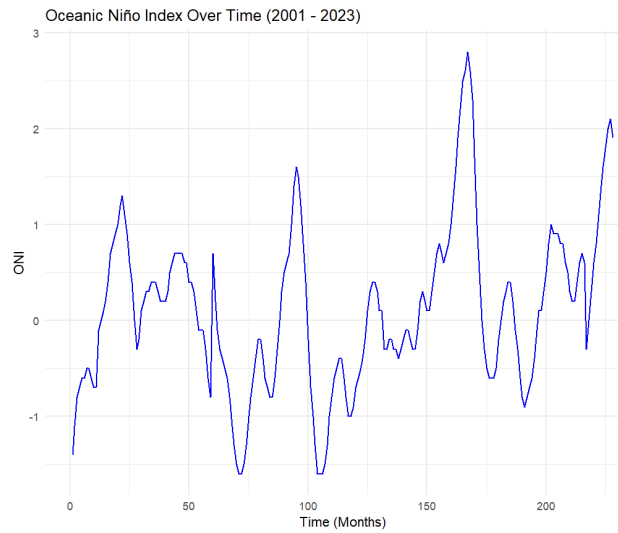


Fig 1, ONI Time Series Line Plot

Local weather variables, including average maximum temperature and total rainfall, also displayed variability across months and years. Temperature and rainfall patterns showed expected seasonal fluctuations, with continuous increases and decreases throughout the year.

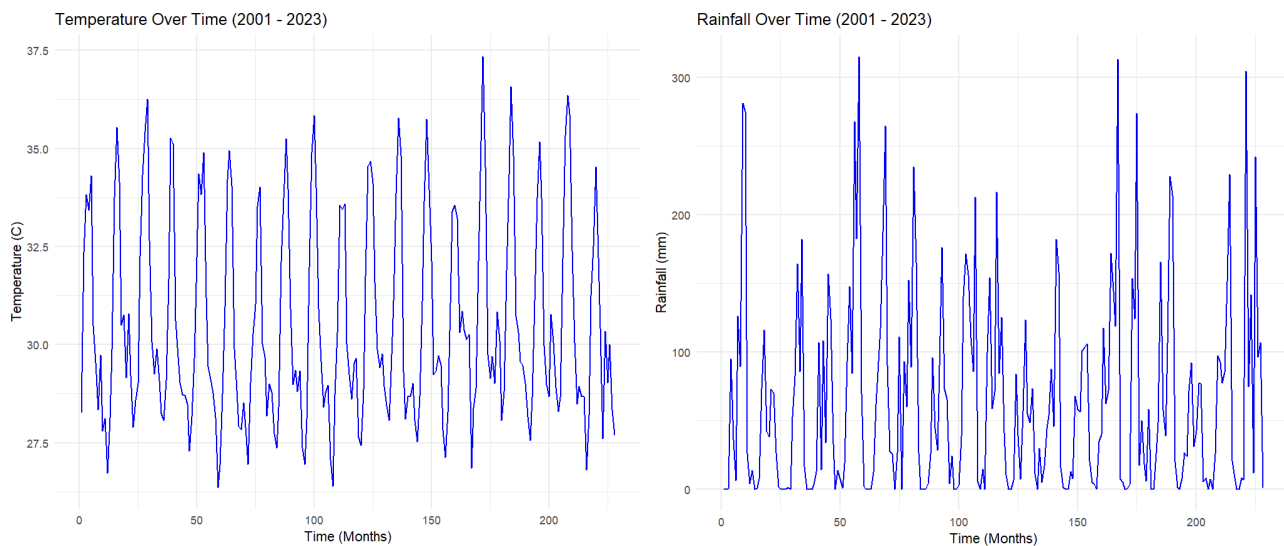


Fig 2, 3 Temperature and Rainfall Time Series Line Plots

Overall, both large-scale climate variability and local weather conditions varied throughout the dataset. Mortality outcomes may be influenced by multiple factors, hence, multiple linear regression was selected to evaluate associations with mortality rates.

Multiple Regression Model

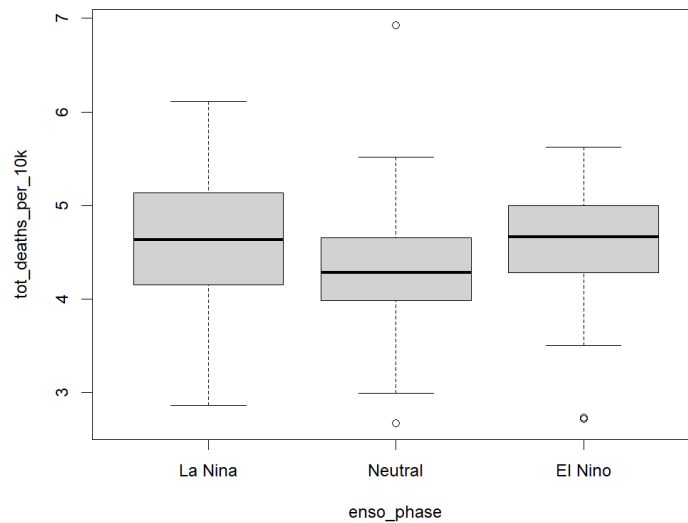


Fig 4, Box plots: ENSO phase vs Mortality

Box plots were used to explain the stronger correlation between mortality and phases of ENSO rather than the continuous ONI values alone. Greater mortality is observed during El Nino and La Nina conditions than during neutral conditions. As a result, ONI values were grouped into ENSO phases(Neutral, El Niño and La Niña) before building the regression model.

First a variance test using anova was run on the enso_phase variable and a p-value of 0.00553 was obtained, confirming statistical significance. This means enso_phase is an important variable to use in the model. Furthermore, local weather conditions like total_rainfall and avg_monthly_temp are also considered as they may be likely to influence the model. The fluctuation in variables seen during the EDA suggests seasonality may also be of importance so that is considered in the preliminary model as well.

Now the regression model looks like this:

$$tot_deaths_per_10k \sim avg_max_temp + total_rain + factor(month) + factor(enso_phase)$$

However, to adjust the model to account for only the significant variables, the summary statistics for the regression model must be considered. Analysing the summary statistics, the statistically significant variables are La Niña with a p-value of 0.0233 and El Niño with a p-value of 0.0429.

Since several variables are insignificant, we need to choose fewer predictor variables. Stepwise AIC was run on the original model.

Reduced multiple regression model:

$$tot_deaths_per_10k \sim avg_max_temp + factor(enso_phase)$$

Discussion

The multiple regression model indicates an association between ENSO phases and mortality. This suggests that large-scale climate variability may have a stronger correlation to mortality rates than local weather conditions.

However, the adjusted R^2 value on the multiple regression model is 0.07246 which is considerably low and does not capture a large extent of the variability associated with the model. This means that the model does not explain mortality perfectly. This is expected as the data used accounts for all mortality in Bangalore and there are several other variables like access to healthcare, disease prevalence and socioeconomic conditions that may need to be considered.

Although limited in its explanatory power, the study indicates important relationships between climate variability and population health. Further research can expand a similar framework to larger datasets, additional environmental variables and geographic locations.

References

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